Introduction to

Elastic Observability

elastic

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    "title": "Pr. Solutions Architect",
    "email": "frederic.maussion@elastic.co",
    "country_name": "France",
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Intro to Elastic Observability

Hands-on !

elastic | The Search AI Company

September 2025

Agenda

101 Elastic 101 & Observability Overview

1 Host Monitoring

2 Kubernetes Monitoring

3 APM

4 AgenticOps

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Elastic — The Search AI Company

Elastic helps everyone transform data into **answers**, **actions**, and **outcomes** with Search AI.

Founded in
2012

5B+
downloads

4,000+
employees

Used by over **54%**
of the Fortune 500

40+
Countries with employees

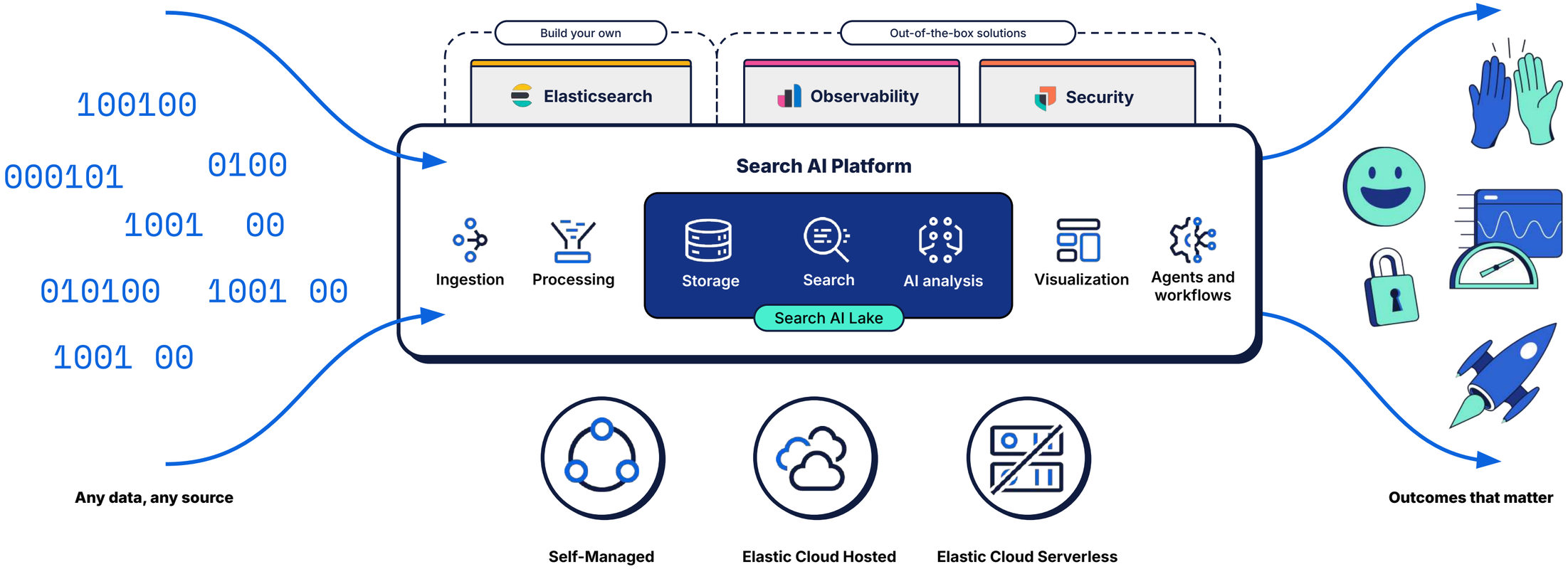
Publicly traded under
ESTC in the NYSE

1,500+
Partners globally

450+
Turnkey integrations

Data as of Mar 2025

Get 10x more value out of your **underutilized data** with **Search AI**



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Elastic Observability Overview

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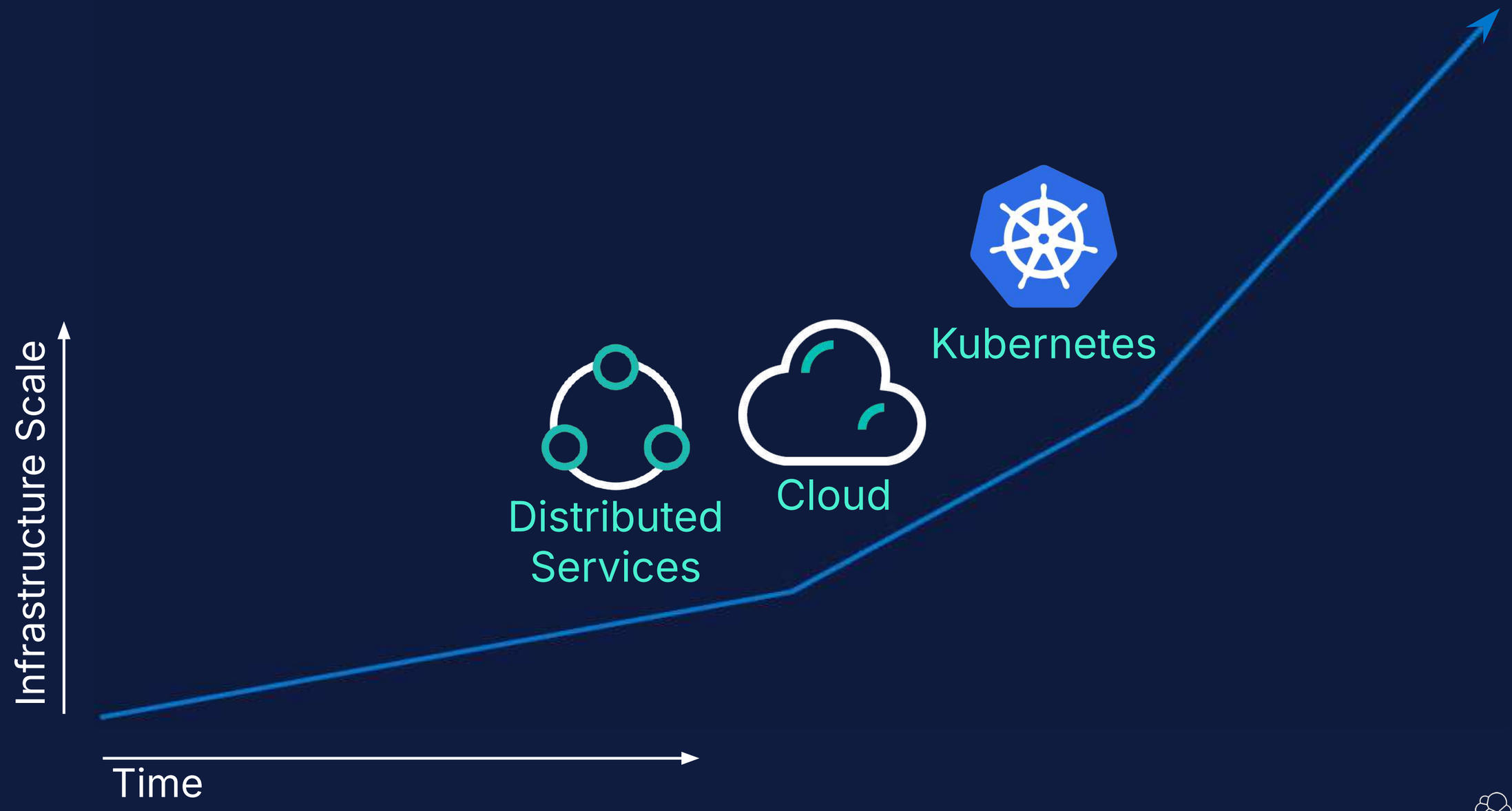
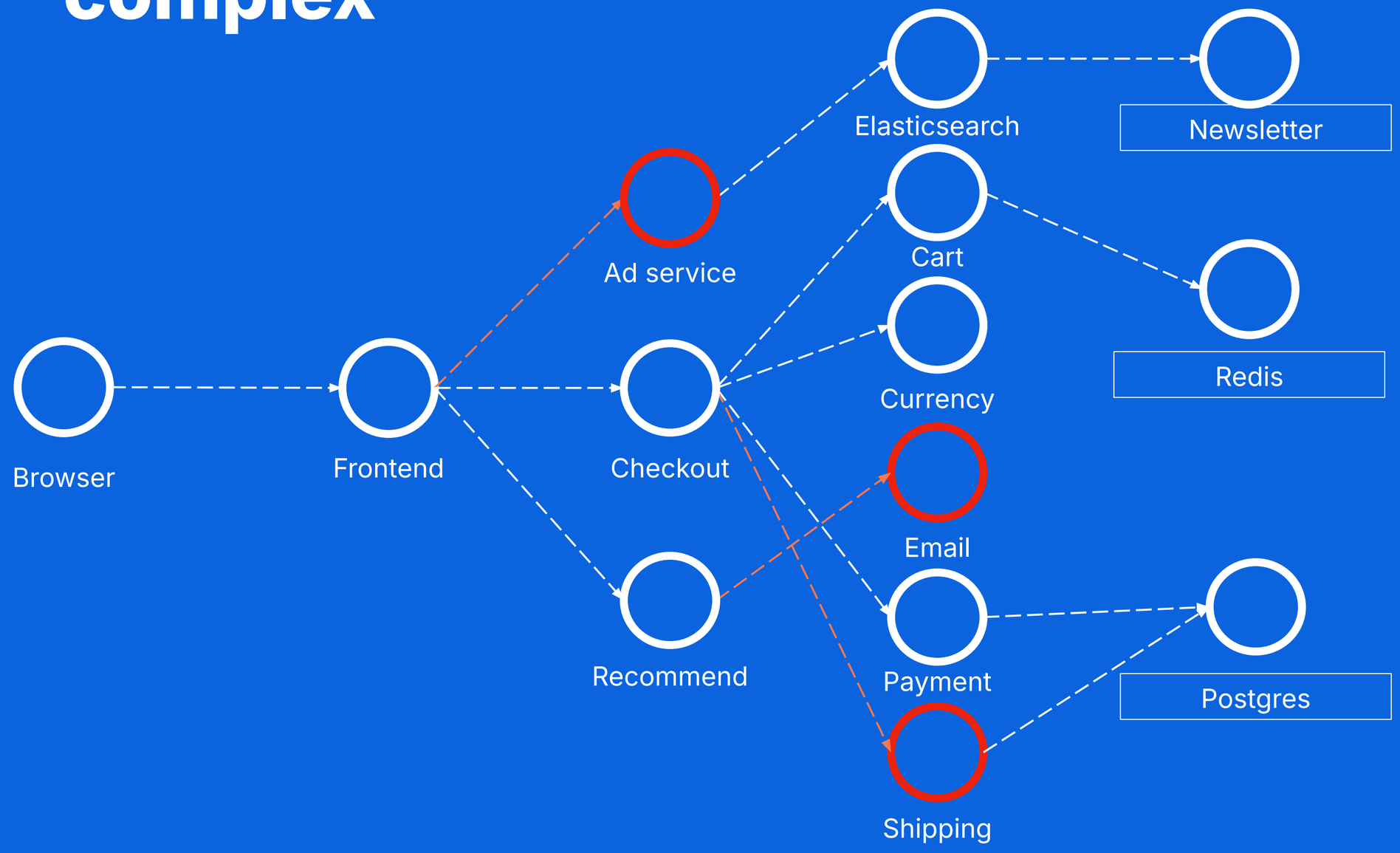
Over the last decade, the business demands of IT have outpaced human reach

So we built systems to scale



Now your applications are complex



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And you can’t get the answers you need

What is my system doing?

Is it working or not?

SRE/DevOps

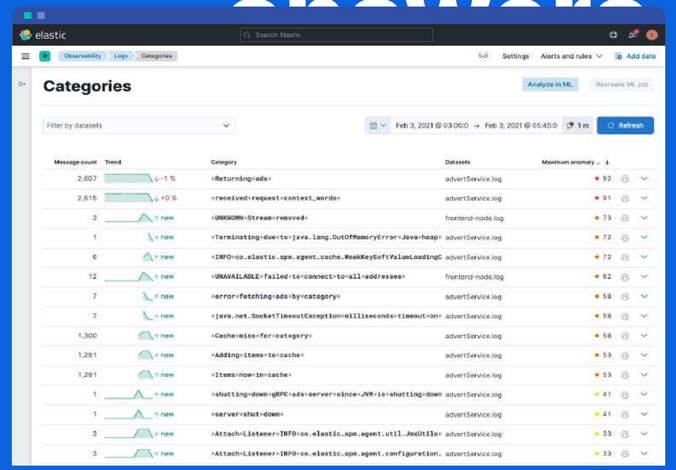
How do I fix it?

SRE/Dev

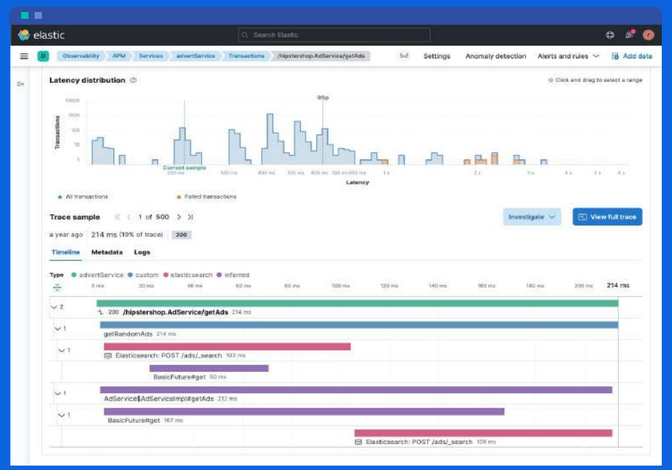
Who is impacted?

SRE/CIO

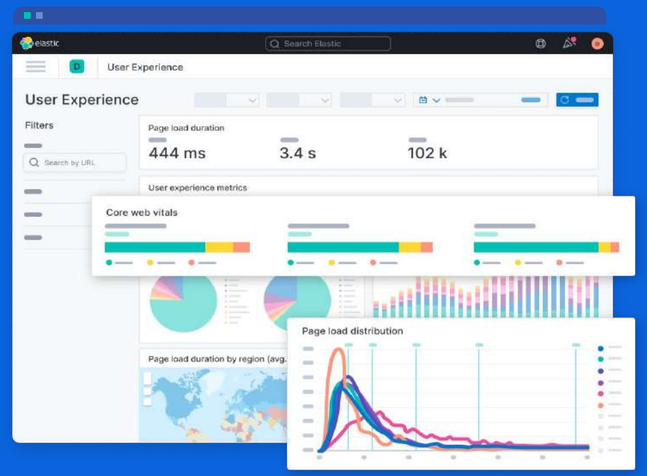
You need signals to get answers



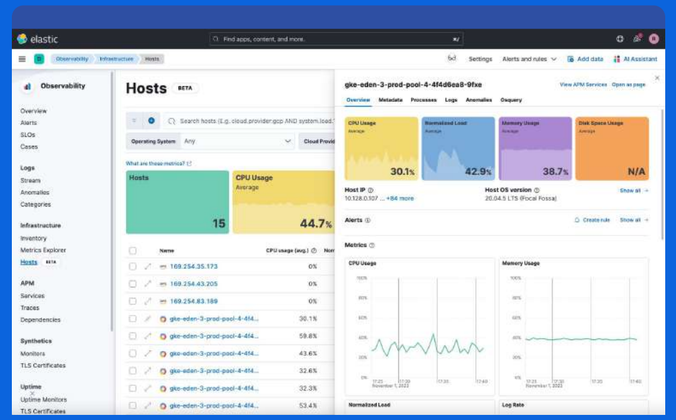
Logging



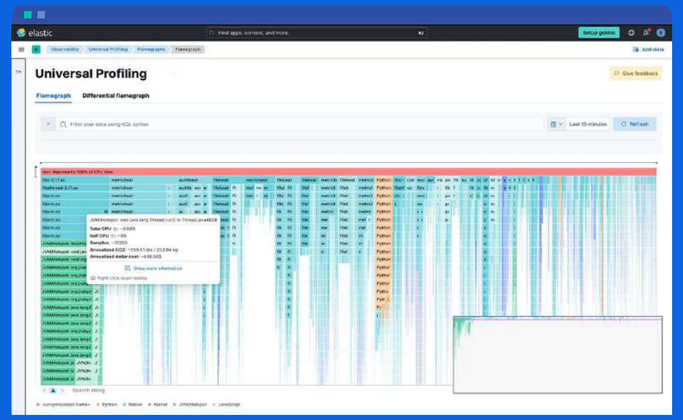
Tracing



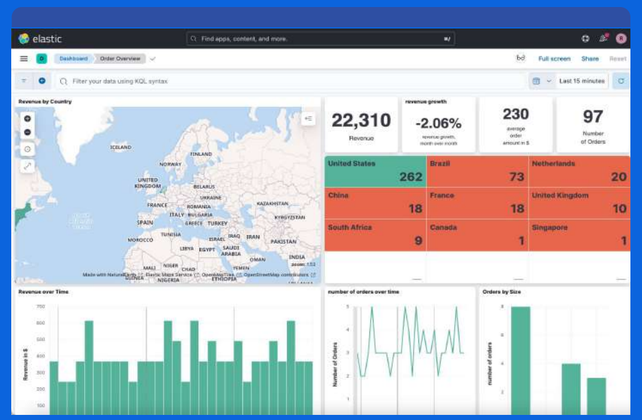
RUM



Infra metrics



Profiling



KPI metrics

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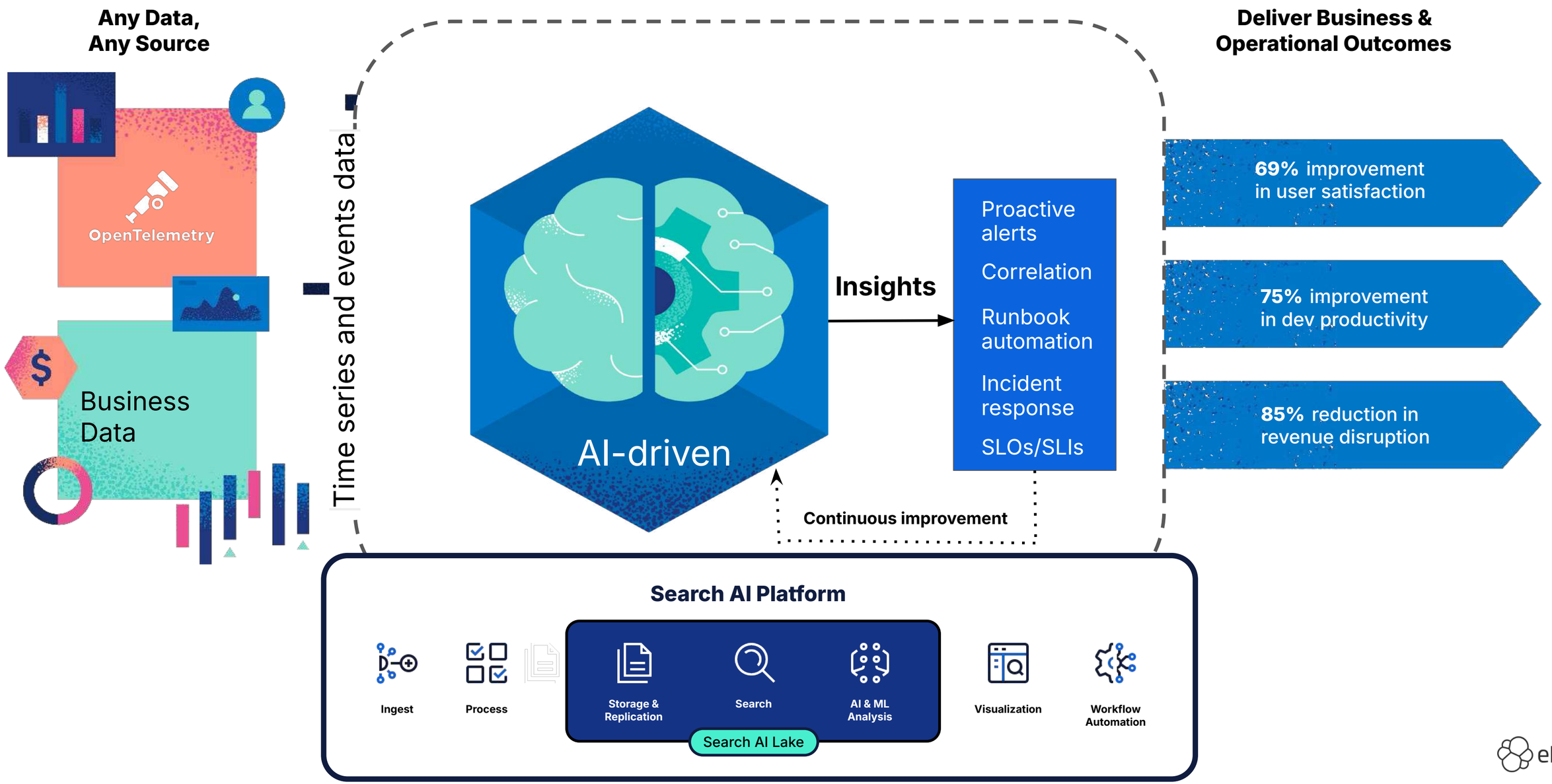
Meet Elastic Observability



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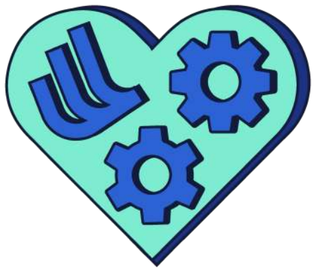
When it comes to minimizing downtime, optimizing performance, and resolving incidents, **you need more** than just swivel chair tools ...



Monitoring - Legacy

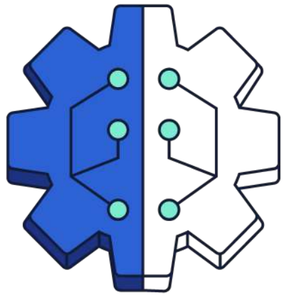
Health monitoring

- Focus on health visibility
- Monitoring “known unknowns”
- Alerting engines
- Manual processes for triage

Observability 1.0 - Current

Siloed observability tools

- Added focus on triage and investigations
- High emphasis on instrumentation
- Black-box tools for apps and infra
- Multi-signal support — as an afterthought

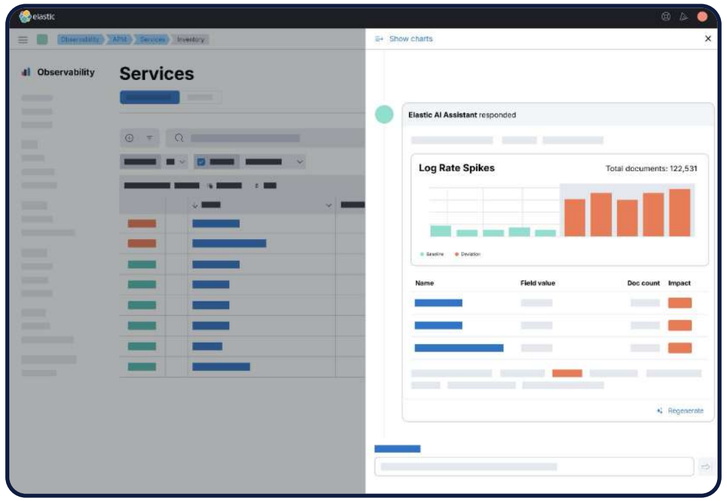


Observability 2.0 - Future

AI and OpenTelemetry

- Solid foundation in enriched logs
- Low instrumentation w/ OTel and AI
- Focus on “unknown unknowns”
- AI-driven investigations and insights

Search AI is the foundation for **next generation observability**



AI Assisted

AI Assistant

Human Assisted

Automatic Root Cause Analysis

Fully Agentic

Trusted Observability Autopilot

Open and flexible: Any data, any source

Integrate with the technology ecosystem you rely on

Data architecture based on OpenTelemetry

Common data model **(OTel + ECS)**

API-driven,
450+ integrations

Open ML models,
plus bring your own

Future proof your investments

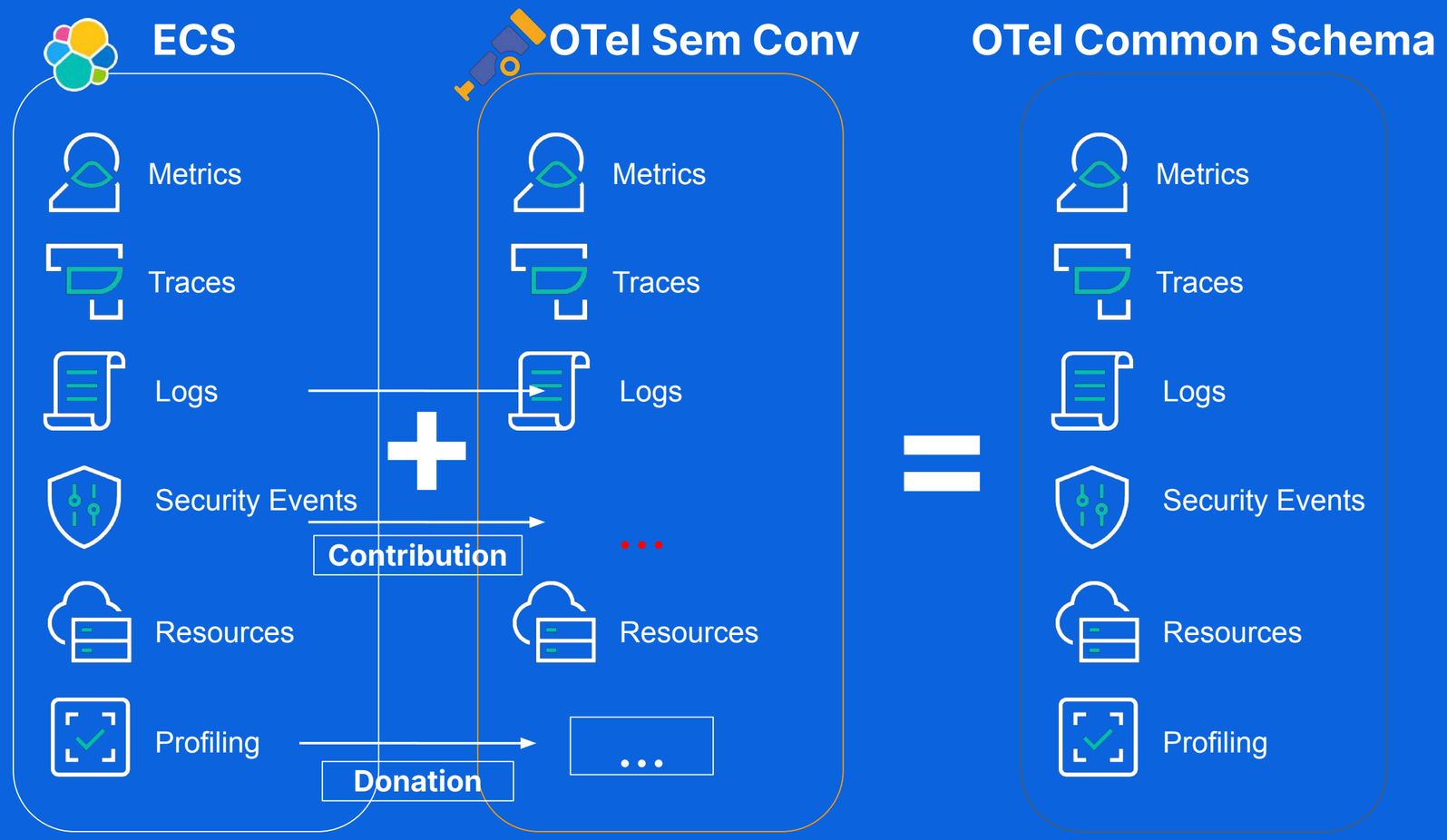
Data Standardisation With Common Schema

Searching *without* ECS/SemConv

```
src:10.42.42.42
OR client_ip:10.42.42.42
OR apache2.access.remote_ip:
   10.42.42.42
OR context.user.ip:10.42.42.42
OR src_ip:10.42.42.42
```

Searching with ECS/SemConv

```
source.ip:10.42.42.42
```



Currently	Now & In Progress

From ingest to insights

Everything emits a signal

Application

Systems

Network

Cloud

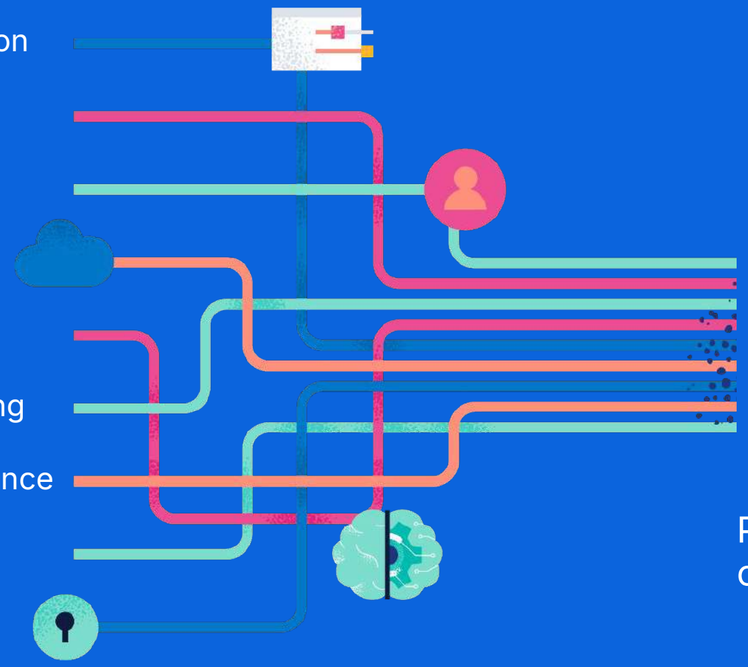
IoT

Debugging

Performance

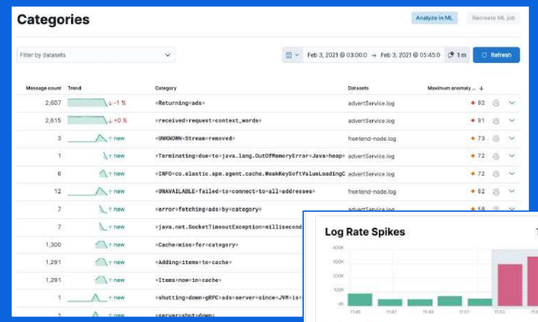
CI/CD

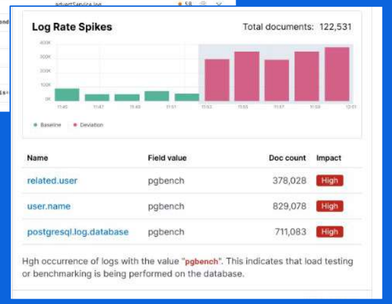
Security



Process billions of logs

AIOPS: AI & analytics





Comprehensive insights

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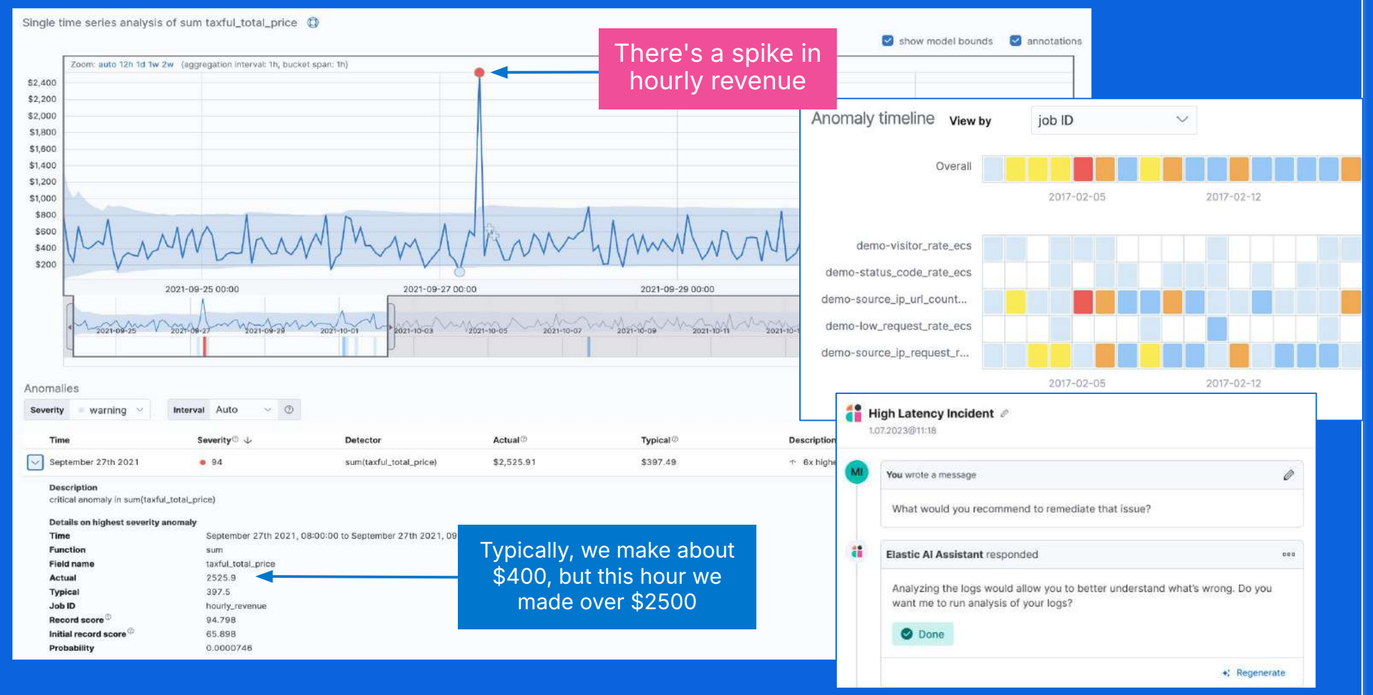
AI & Analytics: Correlate any type of data

Accelerate problem resolution and improve predictability

Detect → **Analyze** → **Remediate**

- Anomaly detection & correlations
- Log categorization, Log rate analysis
- 100+ out-of-the-box customizable ML models
- Generative AI powered

Democratize data and analytics



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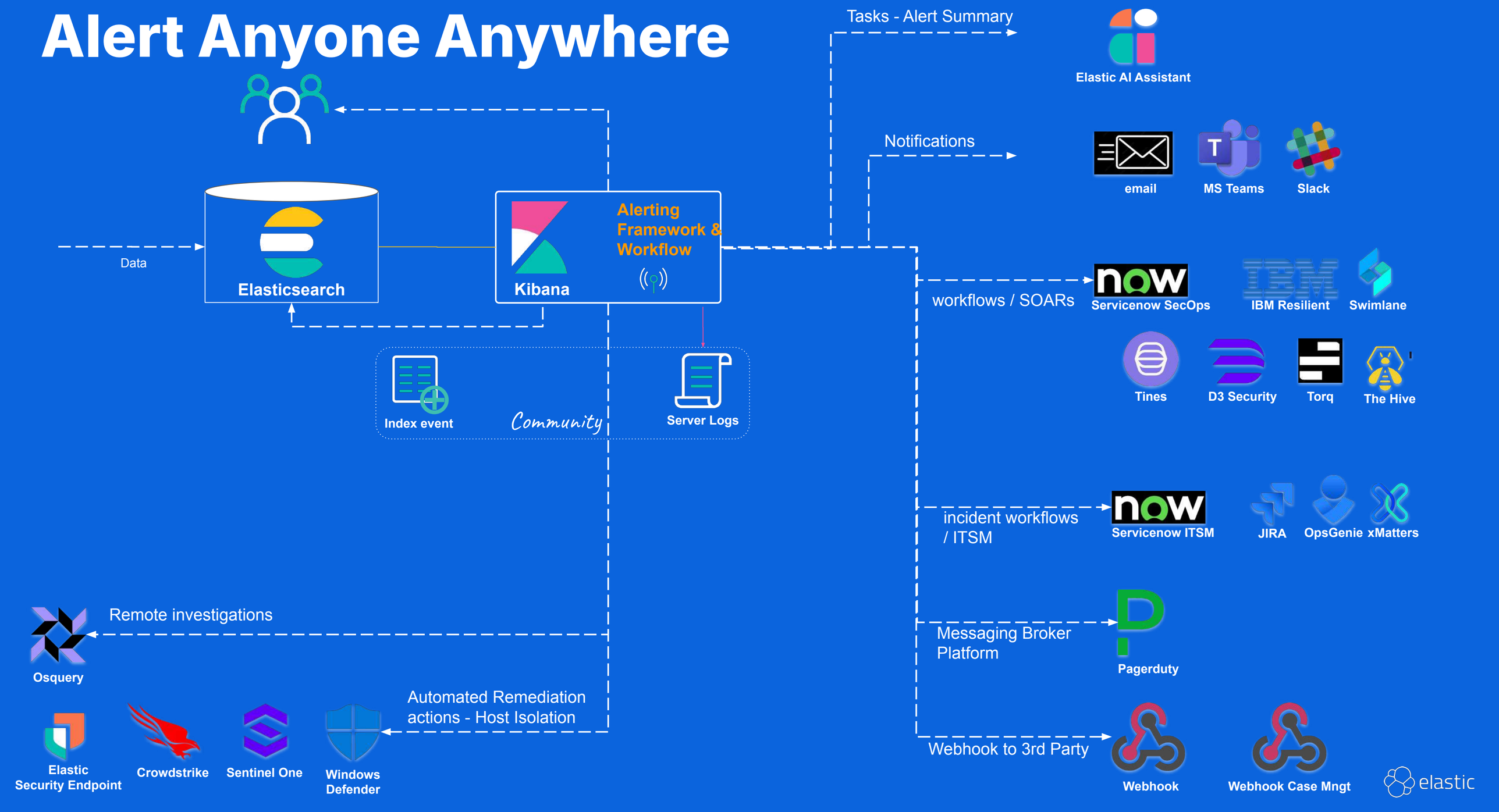
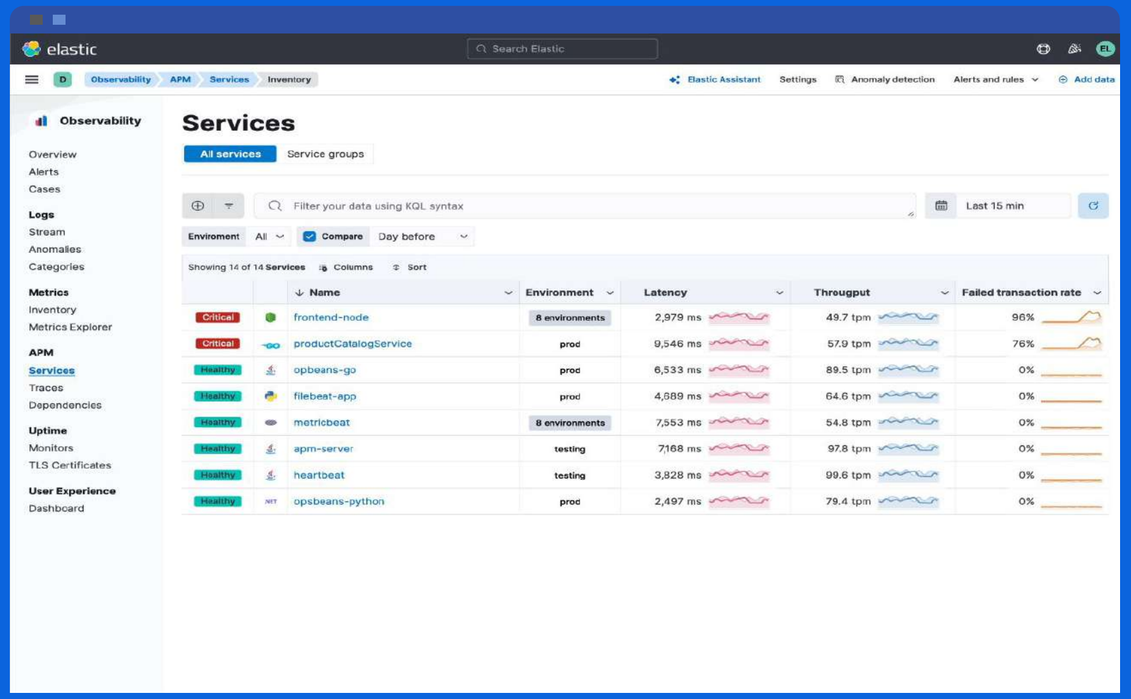
Alert Anyone Anywhere



Elastic AgenticOps for Observability

Detect → **Analyze** → **Remediate**

- Provide Alerting RCA
- Explain log messages
- Explain host processes
- Explain APM errors
- Profiling to analyze libraries and functions
- Visualize metrics
- Use natural language questions for meaningful responses and actions with LLM or internal private data (runbooks, service issues, etc.)



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GitHub

Product issues, runbooks, customer issues, etc.

LLMs
Azure OpenAI
AWS Bedrock
...

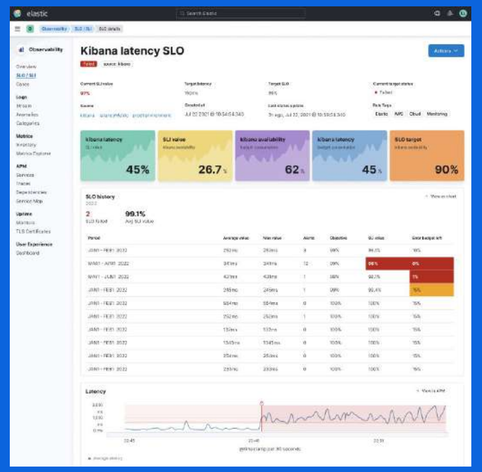
Drive business & operational excellence with SLOs

Get comprehensive insights into complex environments

Business insights	Are you meeting revenue goals?
Front end	Is your customer experience degrading?
Backend services	Have you optimized app performance?
Dev pipeline	How well are CI/CD pipelines working?
Cloud/Infra	Is your infrastructure optimized?

Are you meeting objectives?

SLOs/SLIs



- Track your error budgets
- Ensure production reliability
- Track business objectives

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Labs

1. Host Monitoring
2. Kubernetes Monitoring
3. APM
4. AgenticOps

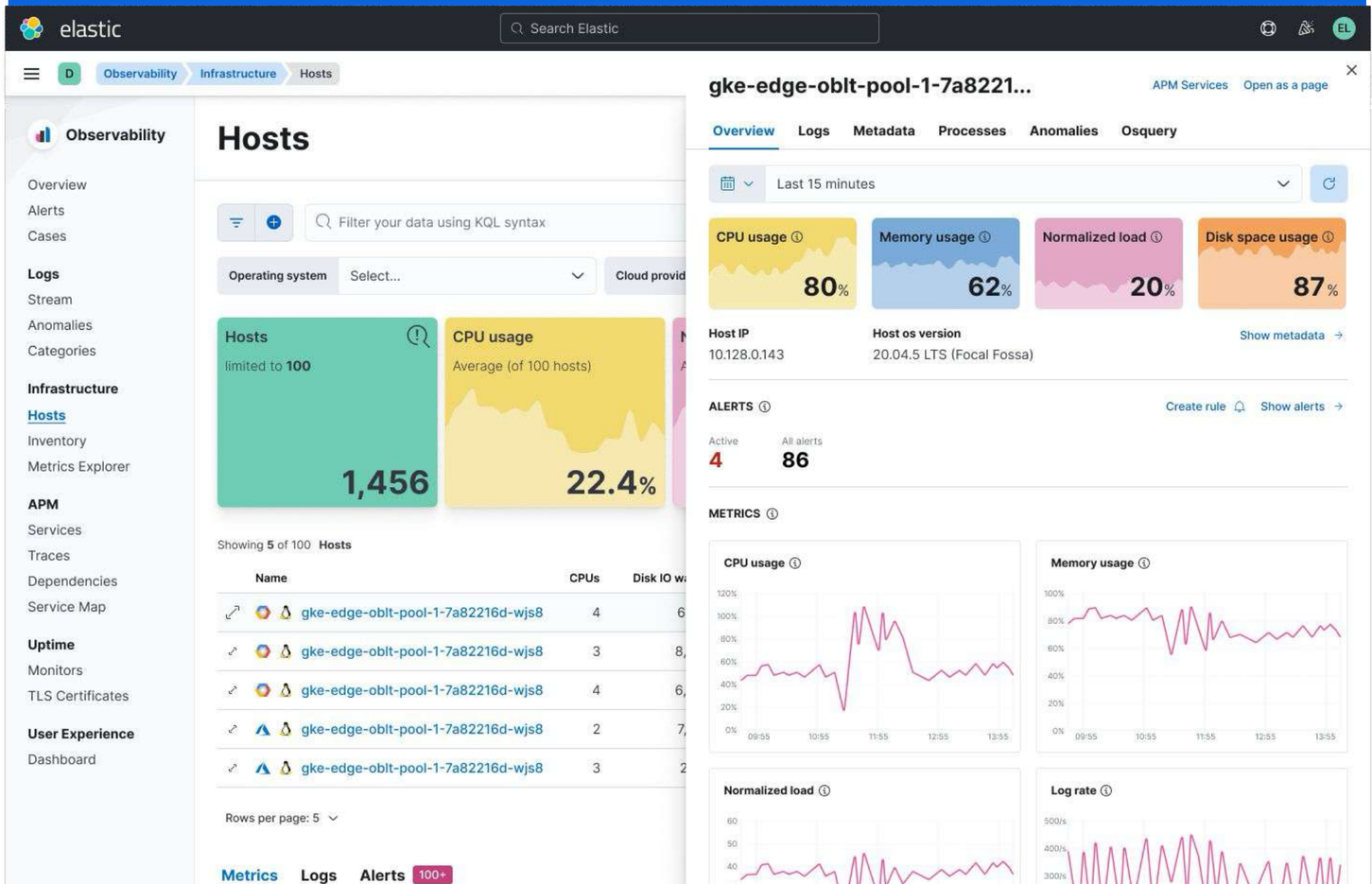
Lab 1 : Host Monitoring Overview

Learn how to use Elastic Observability to monitor your hosts and applications.

This track demonstrates how to:
- **Setup Elastic Agent**
- **monitor system logins**
- **find the root cause of log spikes.**

Challenges:

- **Add Elastic Agent**
- **Investigating System Logins**
- **Quiz - Logins**
- **Add the Nginx Integration**
- **View your Nginx Data**
- **Analyzing a Spike in Log Data**
- **Quiz - Log Spike**
- **Investigating System Logins**
- **Quiz - Log Spike**



Labs

Lab 1
~ 20 minutes

1. Host Monitoring
2. Kubernetes Monitoring
3. APM
4. AgenticOps

20:00

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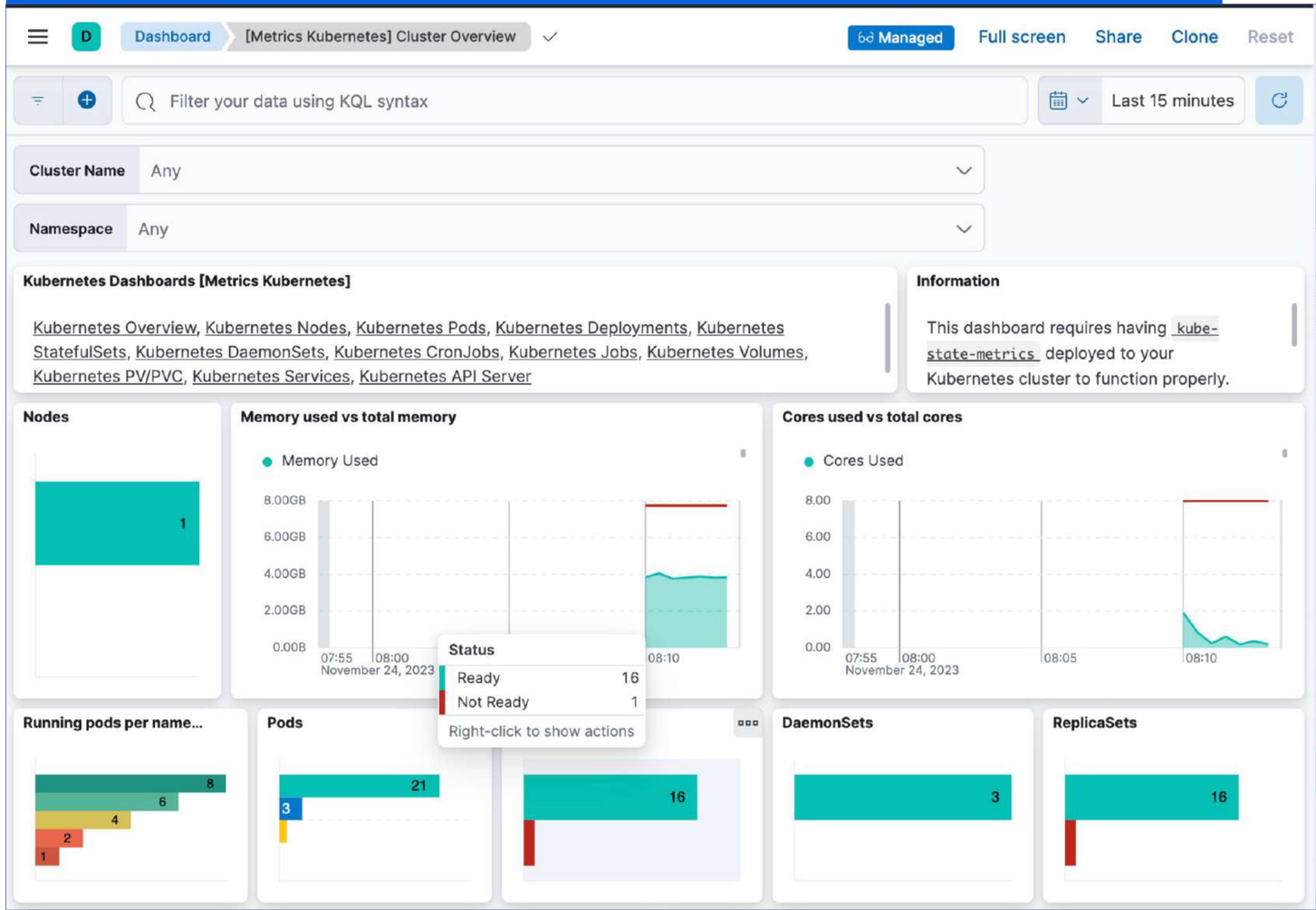
Labs

- ✓ Host Monitoring
- 2 Kubernetes Monitoring
- 3 APM
- 4 AgenticOps

Lab 2 : Metrics - Kubernetes Monitoring

Learn how to use Elastic Observability for Kubernetes monitoring

This track demonstrates how to:
- **Setup Elastic Agent**
- **Collect Logs + Metrics from Kubernetes**
- **Find the root cause of many issues**



Challenges:
- **Elastiflix - our demo application**
- **Add the Docker Integration**
- **Adding kube state metrics**
- **Add the Kubernetes Integration**
- **Troubleshooting Kubernetes Deployment Issues**
- **Quiz - 1**
- **Investigating Further Deployment Issues in Kubernetes**
- **Quiz - 2**

Labs

Lab 2
~ 20 minutes

1. Host Monitoring
2. Kubernetes Monitoring
3. APM
4. AgenticOps

20:00

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Labs

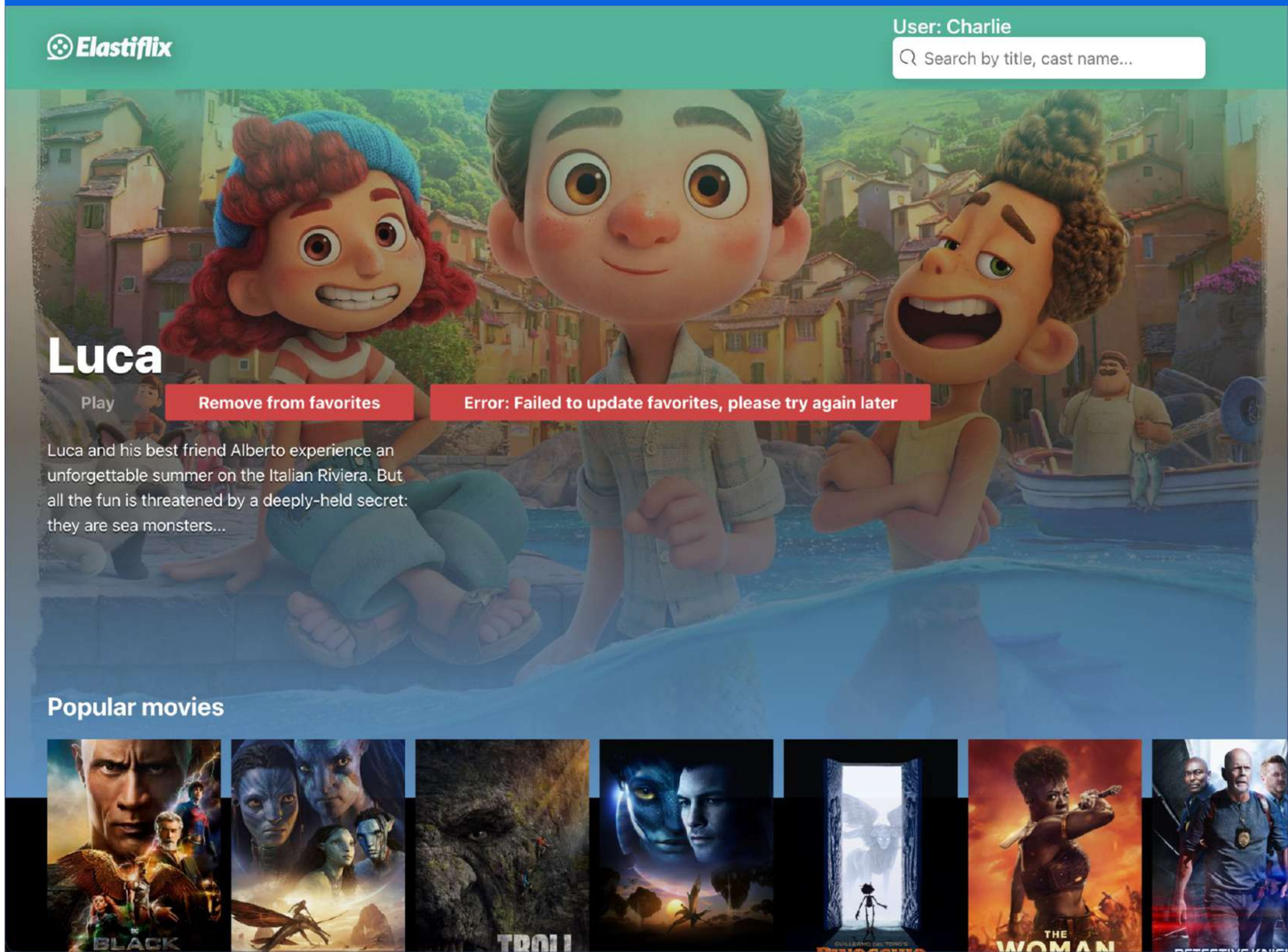
- ✓ Host Monitoring
- ✓ Kubernetes Monitoring
- 3 APM
- 4 AgenticOps

Lab 3 : APM

Learn how to use Elastic Observability for APM.

This track demonstrates how to:
- **Use APM to gain additional insight**
- **Troubleshoot errors with APM**
- **Finding correlations in your data**

Challenges:
- **Elastiflix - our demo application**
- **Elastic APM**
- **A broken application - 1**
- **Quiz - 1**
- **A healthy application - 1**
- **A broken application - 2**
- **Quiz - 2**
- **A healthy application - 2**
- **A broken application - 3**
- **Quiz - 3**



Labs

Lab 3
~ 20 minutes

1. Host Monitoring
2. Kubernetes Monitoring
3. APM
4. AgenticOps

20:00

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Labs

- ✓ Host Monitoring
- ✓ Kubernetes Monitoring
- ✓ APM
- 4 AgenticOps

AgenticOps
Elastic Observability

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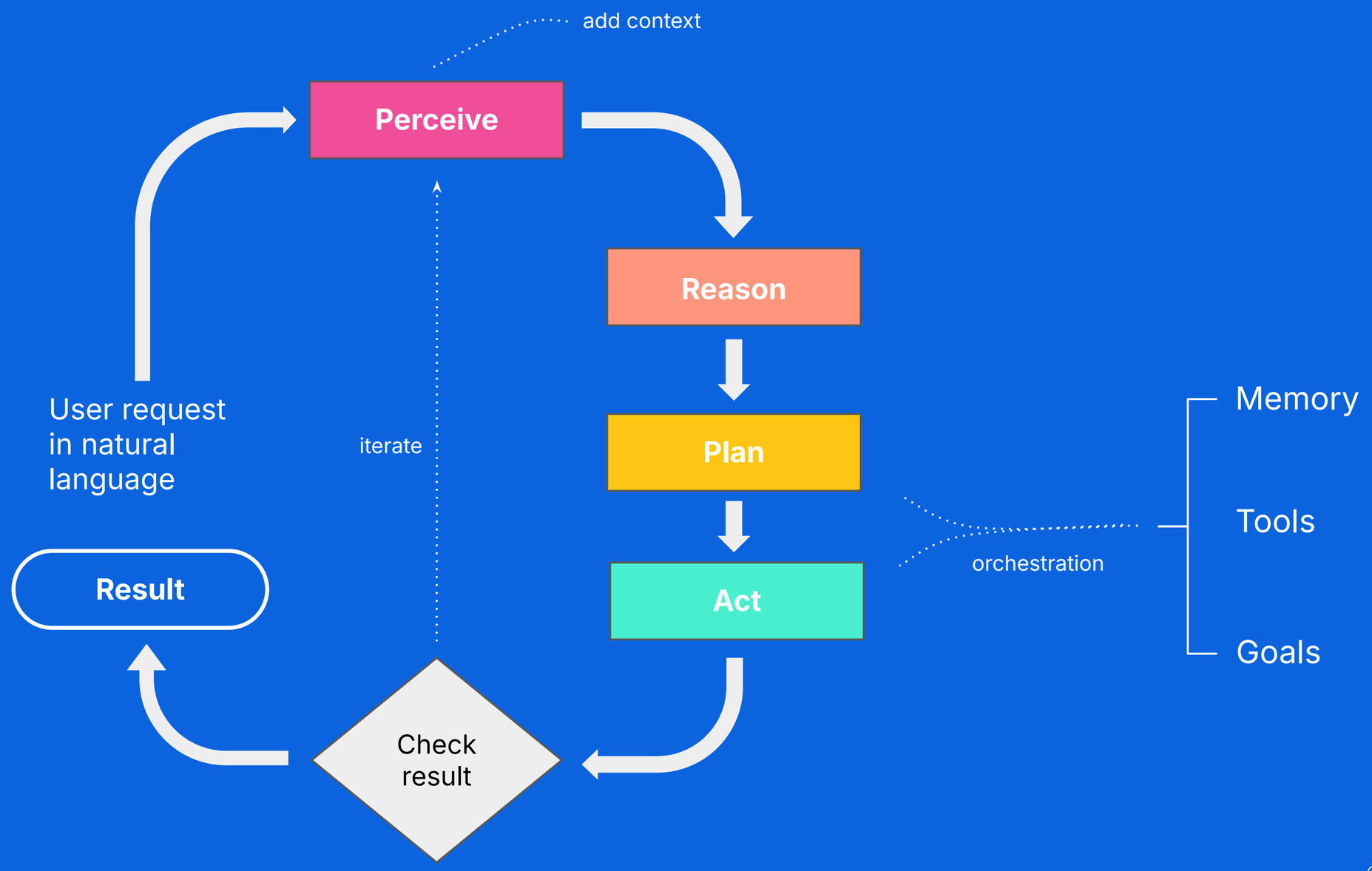
September 2025

An AI agent is a system that can perceive its environment, reason about what it observes, plan a sequence of actions, and execute those actions to achieve specific goals

An agent runs tools in a loop to achieve a goal



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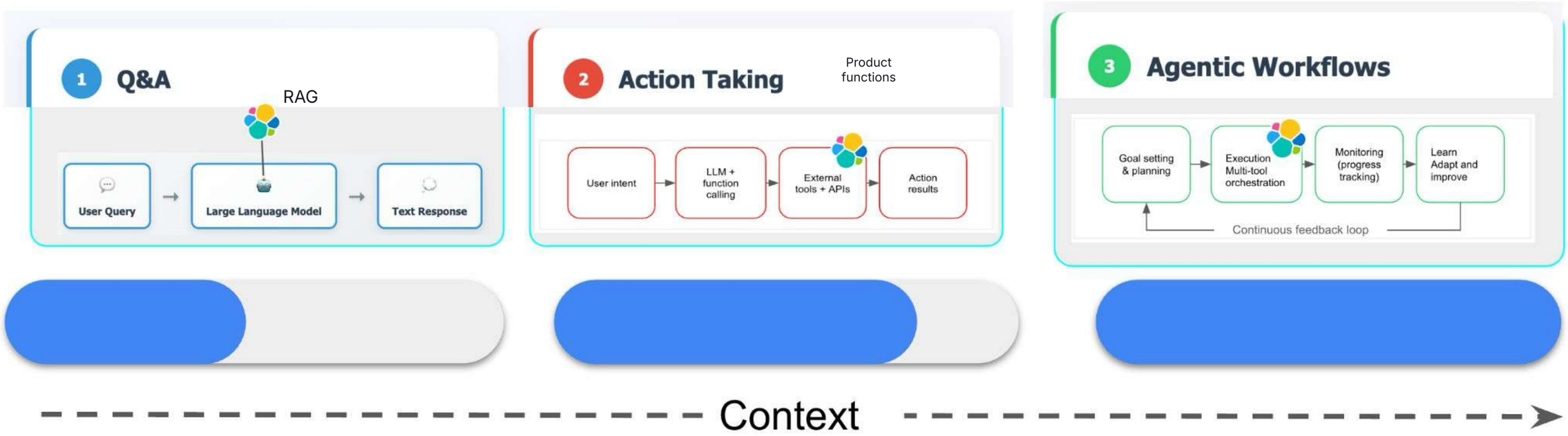
Moving from conversational to agentic AI

Agentic AI doesn’t just answer questions, it takes action



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Agentic AI for SREs

Natural language to action – AI Assistant can access native product actions to create visualizations, dashboards, run ML jobs, reduce alert noise, etc.

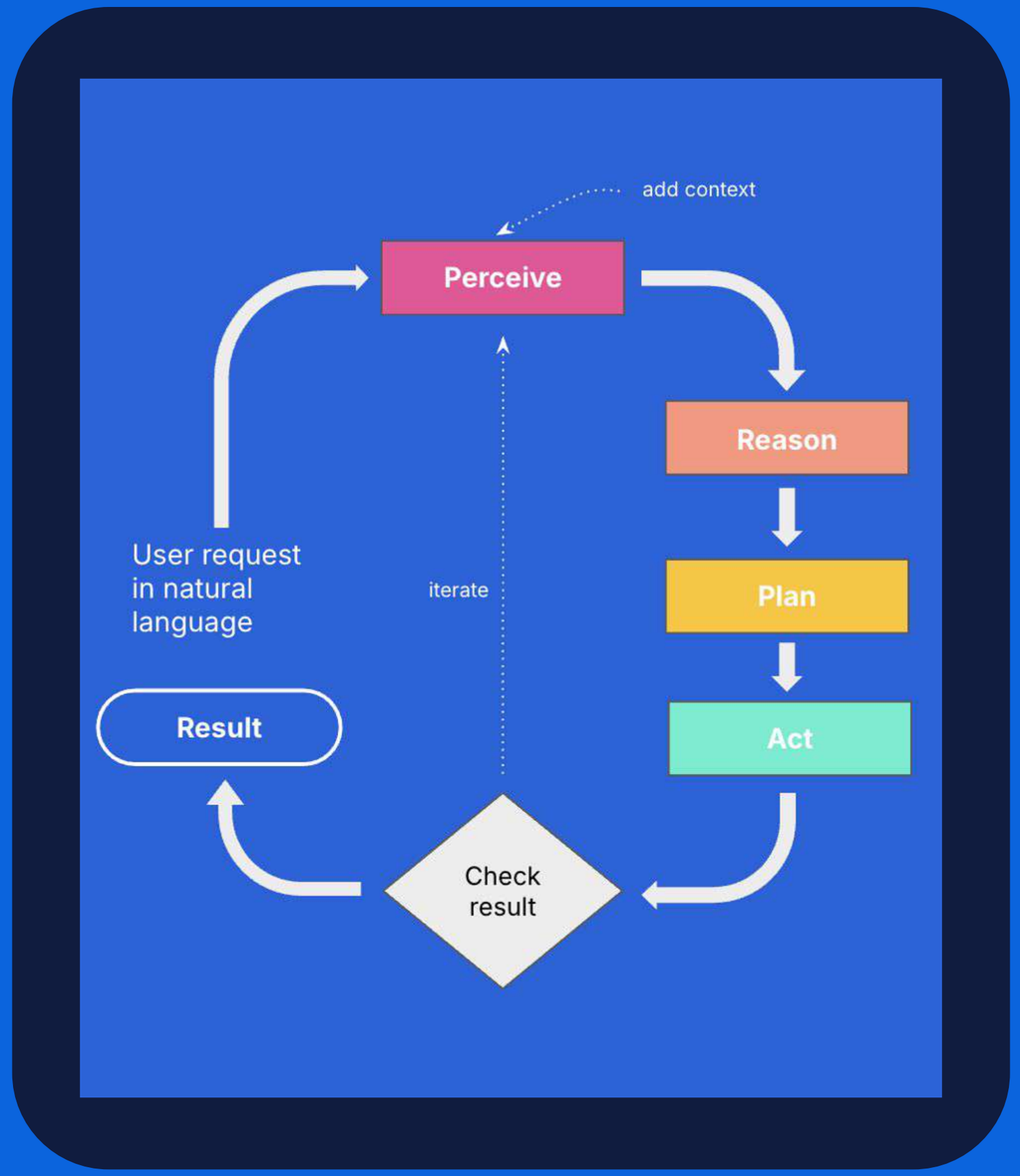
Advanced tasks – bespoke tools for generating complex queries from SRE-supplied use cases, building custom integrations, and more

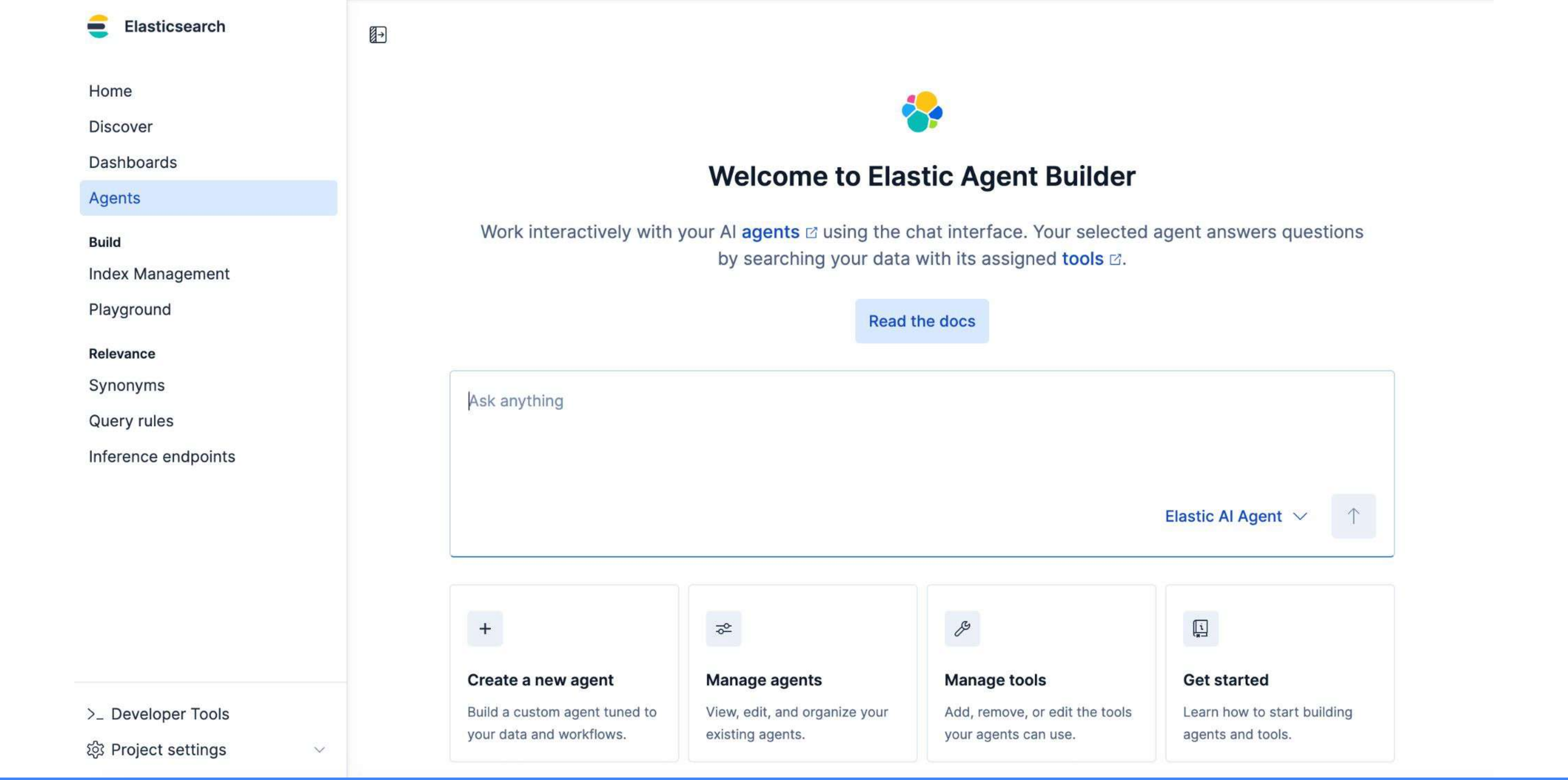
Integrated directly in product – no switching tools or context, everything runs inside Elastic

Transparency – SREs can inspect and refine workflows with AI that shows its work

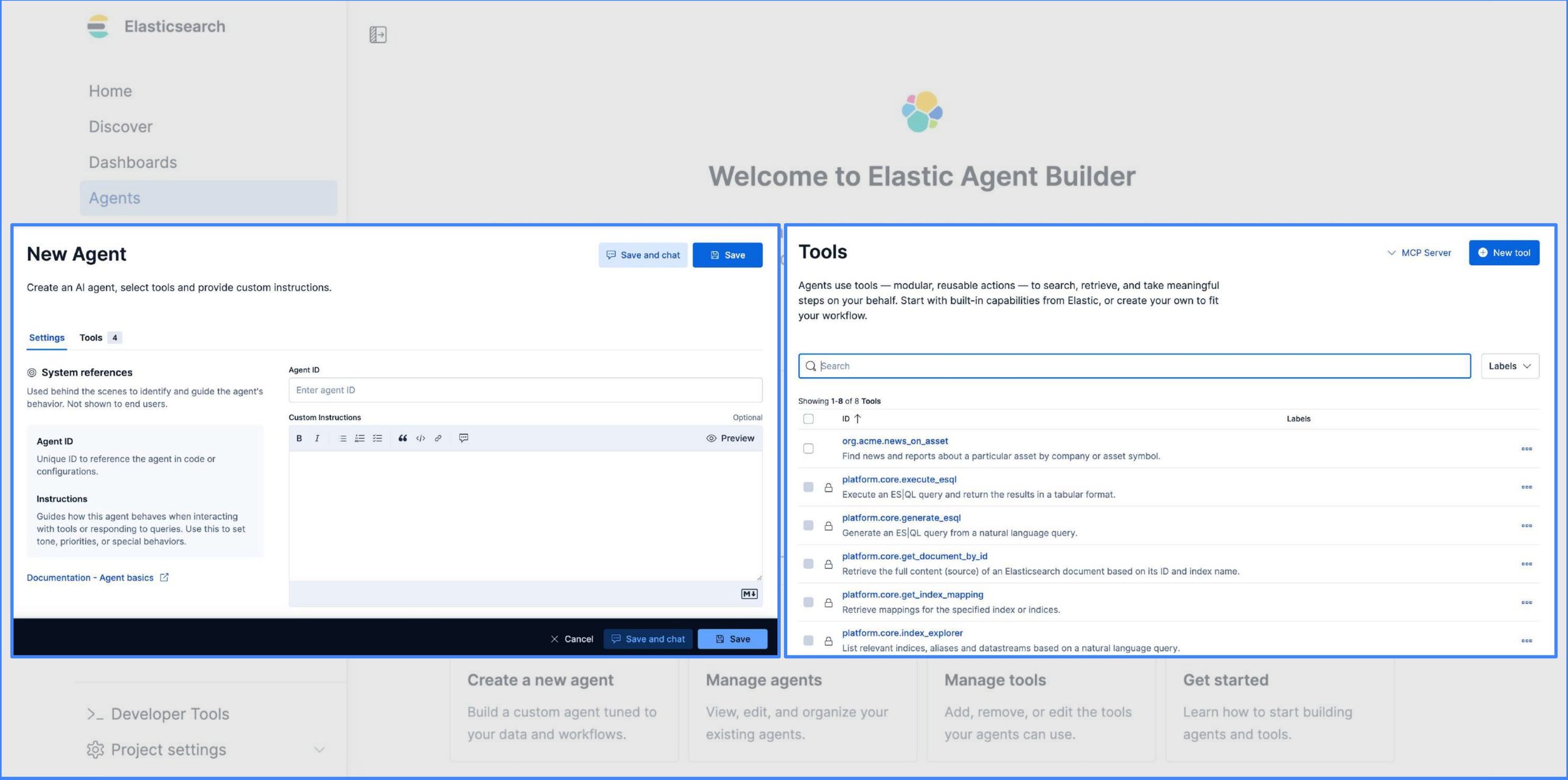
Human-in-the-loop – keeps SREs in control

Contextually grounded - integrates org knowledge sources such as runbooks, github, and past incidents.



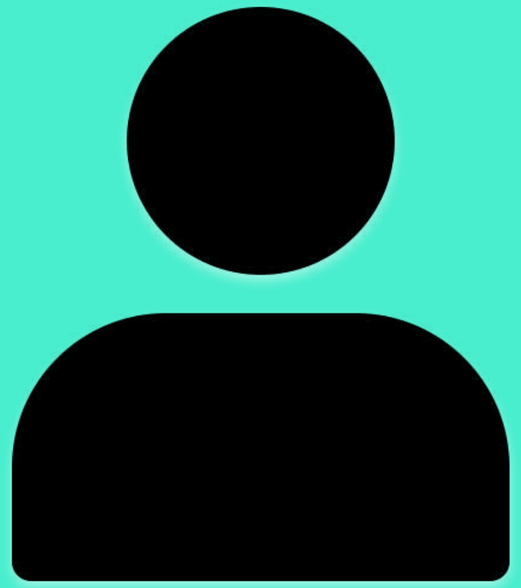


Elastic AI Agent Builder



Elastic AI Agent Builder

Demo #1
Kibana



SRE

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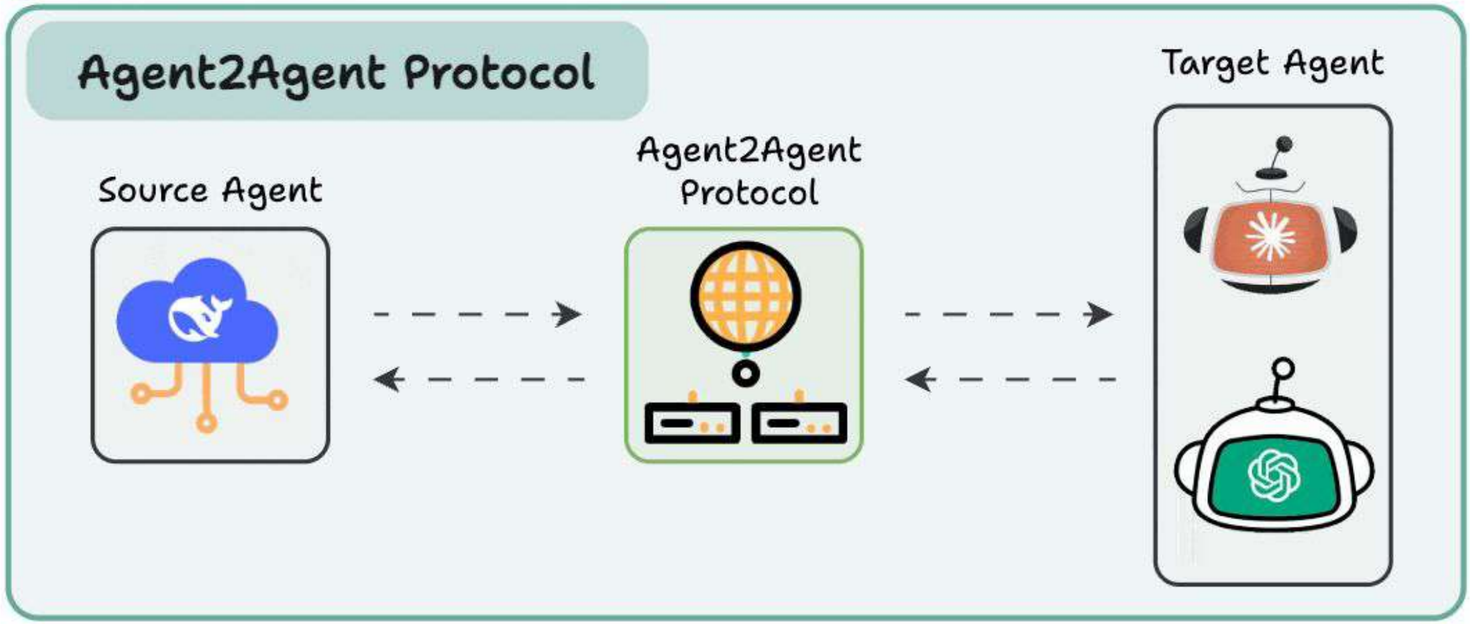
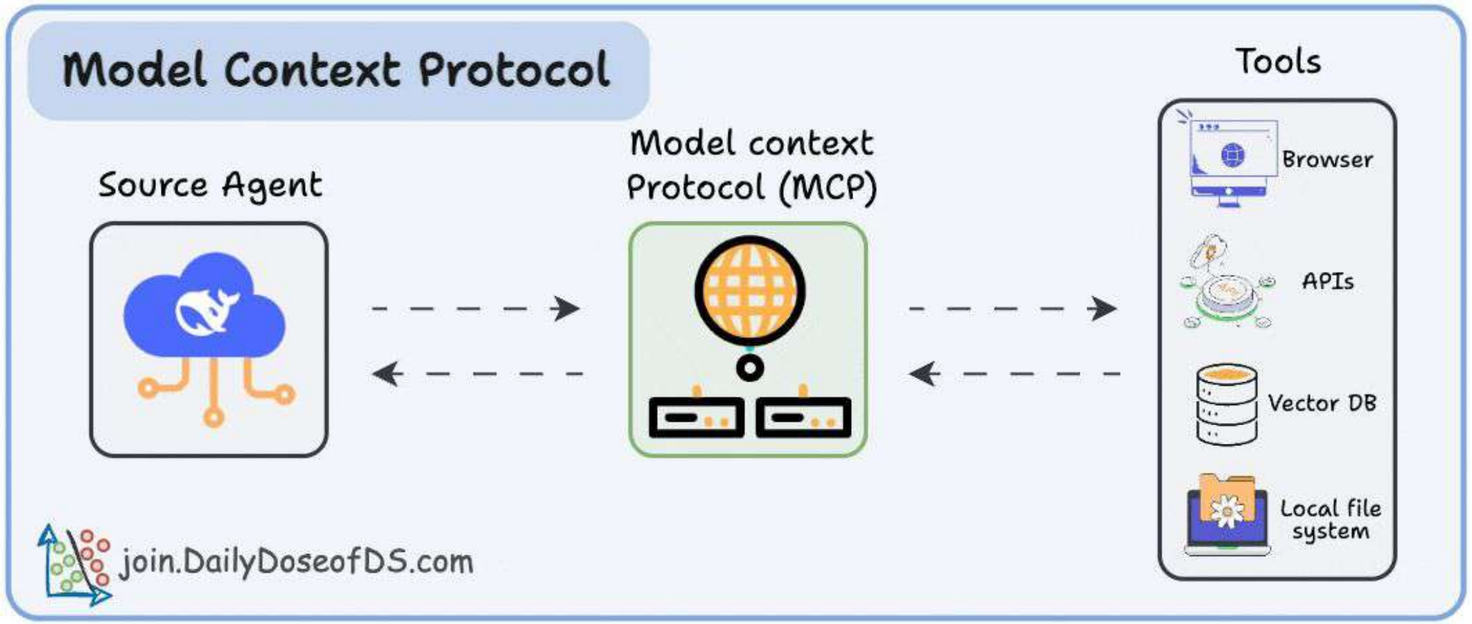
Agent2Agent Protocol vs. Model Context Protocol





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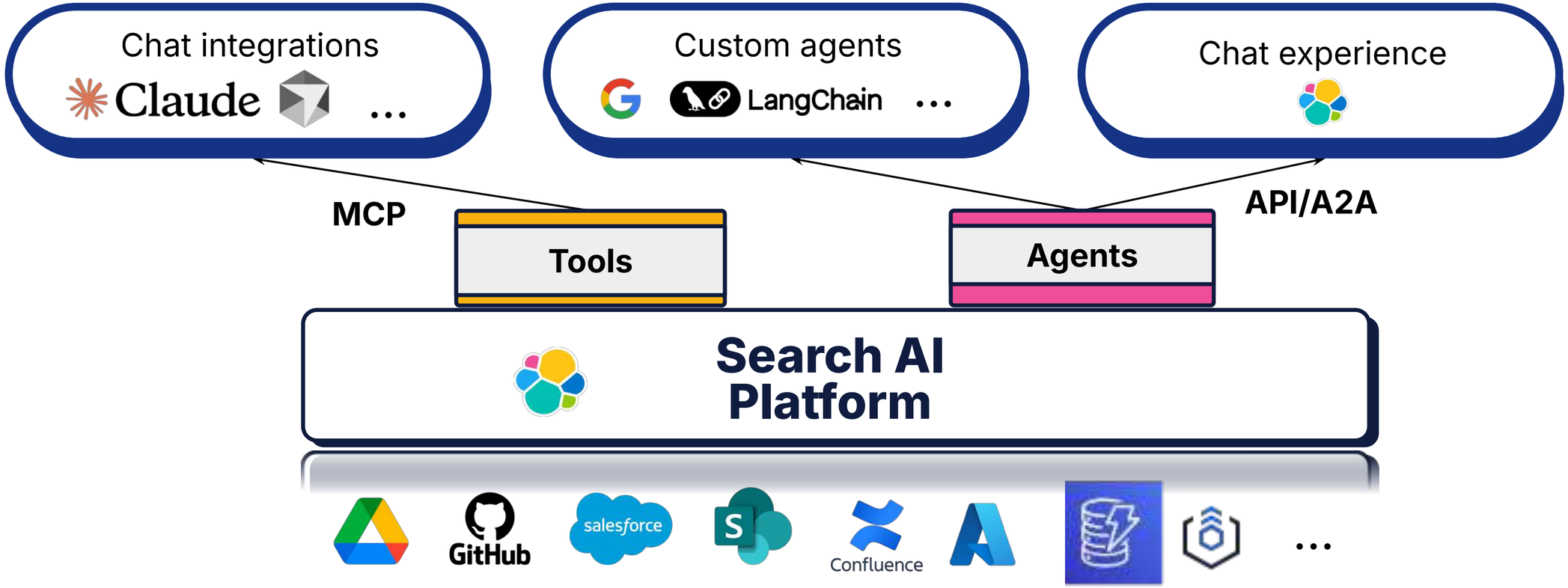
Elastic AI Agent Builder makes it simple



Demo #2
API, MCP & A2A

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Agent Builder on Elastic's Search AI Platform is

1. **Built in**, not bolted on for faster time to value
2. **Scalable, efficient, and secure** for enterprise production use cases
3. **Open and transparent** by design to fit in any ecosystem

Thank You

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